1. a) let,

time to transfer data in CS mode = x

So,

$$\left(300 + x + 200\right) \times \frac{512}{4} = 12564000$$

$$\Rightarrow \cancel{30}\ 500 + x = 98156.25$$

$$\Rightarrow x = 97656.25 \text{ ns}$$

$$\therefore x = 0.00009765625 \text{ s}$$

So,

In 0.00009765625 s data transferred 4 bytes

$\therefore$ " 1 s " " $\dfrac{4}{0.00009765625}$ bytes

$$= 40960 \text{ bytes}$$

$$= \frac{40960}{1024} \text{ KB}$$

$$= 40 \text{ KB}$$

$\therefore$ data rate = 40 KB/s

b) In burst mode,

$$300 + \frac{8192 \times 10^9}{40 \times 1024} + 200 = 200000500 \text{ ns}$$

In CS mode,

$$\left(300 + \frac{4 \times 10^9}{40 \times 1024} + 200\right) \times \frac{8192}{4} = 201024000 \text{ ns}$$

So, burst mode is faster.

2. a)

$$\begin{array}{r} 8000\text{ H} \\ -\ 0001\text{ H} \\ \hline 7FFF\text{ H} \end{array}$$

0111 1111 1111 1111

CF = 0 ; no carry out

SF = 0 ; MSB of result = 0

PF = 1 ; even number of 1's in lower byte

OF = 1 ; Neg − Pos should be ~~positive~~ negative but result is positive

b) The offset is 16 bit.

So, memory locations in a segment $= 2^{16}$

Size of each location $= 8$ bit
$= 1$ byte

So, size of segment $= (2^{16} \times 1)$ byte

$= 64$ KB

3. a) $SS = 2526H$
$BP = 1105H$

$\therefore$ Physical address $= SS \times 10H + BP$

$= 2526 \times 10H + 1105$

$= 26365H$

1) Segment $= \dfrac{26365 - 3C5A}{10}$

$= \dfrac{2270B}{10} \rightarrow$ fraction

So, invalid.

ii) offset = 26365 − (213A × 10H)

= 4FC5 → valid

iii) CS = A231 H

First physical address = A2310 H

Last " " = A2310 + FFFF

= B230F

3) b)

$$2^{n_A} \times 16 = 128 \times 2^{10} \times 8$$

$$\therefore n_A = \log_2\left(\frac{128 \times 2^{10} \times 8}{16}\right)$$

$$= 16$$

$$2^{17} \times m = 128 \times 2^{10} \times 8$$

$$\Rightarrow m = \frac{128 \times 2^{10} \times 8}{2^{17}}$$

$$= 8$$

4(a)

$\mu p \rightarrow$ AB 20 bit, DB 8 bit

RAM $\rightarrow$ AB 11 bit, DB 8 bit

$$\text{RAM capacity} = \frac{2^{11} \times 8}{2^{10} \times 8} = 2^1 = 2KB$$

memory needed KB

$$\therefore \text{RAM needed} = \left\lceil \frac{8}{2} \right\rceil = 4$$

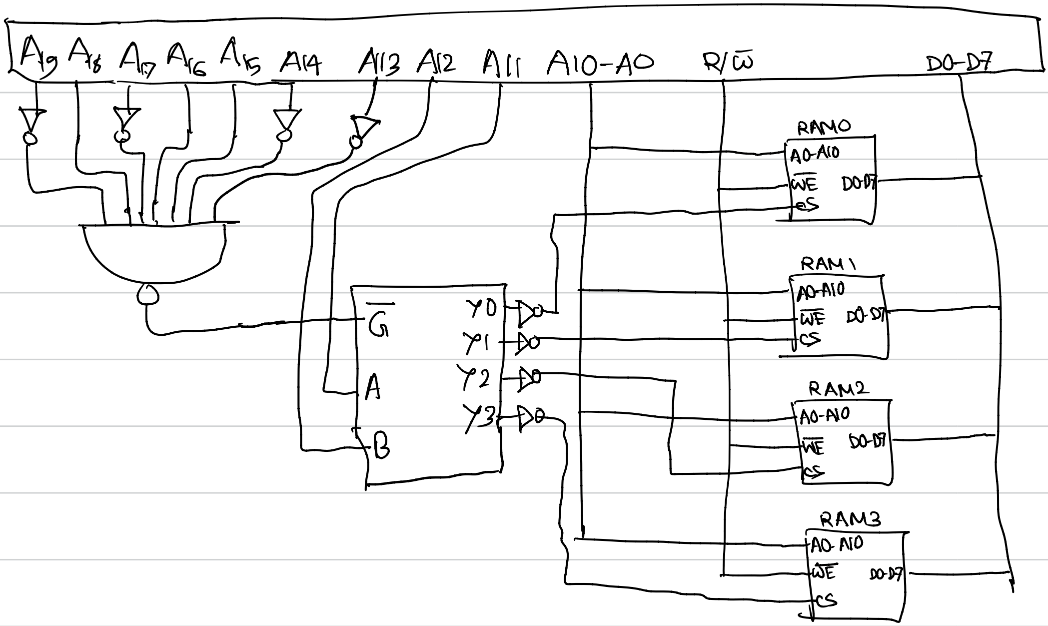
8 4 2 1

$\therefore$ Decoder input line 2

output line $2^2 = 4$

58000 to 59FFF

$\underbrace{0101100}_{A_{19} \text{ to } A_{13}}0$ $\qquad$ $\underbrace{0101100}_{A_{19} \text{ to } A_{13}}1$



A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	A_{14}	A_{13}	A_{12}	A_{11}	$A_{10}-A_0$	RAM	Hex
0	1	0	1	1	0	0	0	0	0....0	0	58000
0	1	0	1	1	0	0	0	0	1....1		587FF
0	1	0	1	1	0	0	0	1	0....0	1	58800
0	1	0	1	1	0	0	0	1	1....1		58FFF
0	1	0	1	1	0	0	1	0	0....0	2	59000
0	1	0	1	1	0	0	1	0	1....1		597FF
0	1	0	1	1	0	0	1	1	0....0	3	59800
0	1	0	1	1	0	0	1	1	1....1		59FFF

5 (a)

motion

Type × 4h = 00368h

∴ type = DAh

∴ INT DA

0036B	CS
0036A	CS
00369	IP
00368	IP

fire alarm

type × 4h = 001E4h

∴ type = 79 h

∴ INT 79h

001E7	CS
001E6	CS
001E5	IP
001E4	IP

79h < DAh

∴ INT 79h will be executed first

(b) 4 × 128 = 512 bytes